4.6 Isometry

Lesson Introduction

Benchmarks	MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.	Learning Targets	• I can describe a sequence of transformations that map one figure onto another congruent figure. • I can draw an image given a preimage and a sequence of transformations on a coordinate plane.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can rigid motions be used to justify that two figures are congruent? • Why do two figures remain congruent once rigid motion is applied?
Lesson Narrative	In this lesson, students begin to look at congruency through the lens of rigid motions. Students will begin by exploring transforming parts using multiple coordinate planes and be introduced to the concept of isometry. They will then continue exploring transformations and isometry in order to bridge that knowledge with the formal definition of congruence in terms of rigid motions.

Component Summary				
	4.6.1 Warm-Up (~5 min.)	Which One Doesn't Belong? *(SE p.* 207)	4.6.3 Exploration (10-15 min.)	Exploring transformations and isometry *(SE p.* 208)
	4.6.2 Exploration (15 min.)	Exploring transforming parts *(SE p.* 207)	4.6.4 Bring It Together (5-10 min.)	Synthesis for formal definition of congruence in terms of rigid motions *(SE p.* 211)
			4.6.5 Wrap-Up (~5 min.)	Growth Mindset survey *(SE p.* 212)

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Isometry • Transformation • Definition of congruence in terms of rigid motions Additional Terms: N/A	• Patty paper • (Optional) Graph paper	N/A

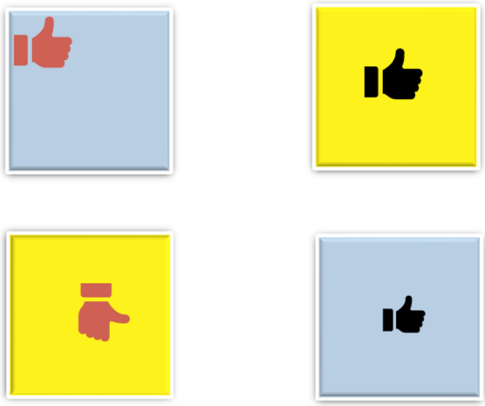
Teacher Tips	Common Misconceptions	Things to Consider
	• Students may not identify parallel arrows during the 4.6.2 Exploration and not preserve elements of distance. • Students may misinterpret the preserving of distance **and** angle measure as either distance **or** angle measure.	• Exploration and discovery-based learning dominates this lesson. Highlight productive struggle and consider providing strategies for working through that struggle, including sentence stems or partner work as embedded scaffolding throughout each exploration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display During 4.6.2 Exploration and 4.6.3 Exploration, students are engaged with a partner or small group. During this time, consider listening for and scribing the output you are hearing from students using written words, diagrams, and pictures. This can then be organized, revoiced, or explicitly connected to other language in a display for all students to use.	**MA.K12.MTR.1.1: Participate in Effortful Learning** Learning can feel like a struggle, and each Exploration provides students the opportunity to flex their productive struggle muscles as they engage in discovery-based learning as they explore and apply working definitions of isometry, congruence, and rigid motion. Incorporate clarification language like "perseverance," "positive mindset," and "community of learners" into your responses throughout the lesson.
Differentiate Instruction	The use of patty paper during each Exploration provides additional entry points for all learners. Combined with the discovery-based approach of this lesson, students are positioned as experts of their own learning experiences regardless of their initial entry points in the process. Consider having students share their different learning experiences as a strategy for collective learning, building confidence, and highlighting flexibility in learning.	
Lesson Notes:		

4.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students will review four given images to determine which one does not belong.*

4.6.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Explain your choice.



Answers will vary. The figure on the bottom right does not belong because the hand in this figure looks smaller than the hands in the other figures. The figure in the top left does not belong because it has the only hand that is located in the corner of the figure. The figure in the bottom left does not belong because it is the only figure where the thumb is not pointing up.

The Warm-Up intentionally positions students to attend to precision in the location, orientation, and other features of each image to better prepare for precision in their 4.6.2 and 4.6.3 Explorations.

After students have chosen one image and explained their responses, challenge them to find at least one other image and provide an explanation for that choice.

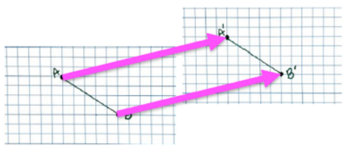
4.6.2 Exploration: Transforming Parts

Component Overview: *Students will be doing an exploration with transforming parts. For this activity, students should be working with a partner or small group to go through each of the steps in the process of this activity. At the conclusion of this activity, students will formalize the definition of isometry.*

4.6.2 Exploration: Transforming Parts

Provide space and opportunity for productive struggle throughout this Exploration. Students will engage in meaningful learning opportunities by making mistakes or identifying what information they need to know and formulating what questions to ask.

1. Traci drew $\overline{AB}$ on a piece of graph paper. She then moved the graph paper up and over to the right, as shown by the arrows. The movement created $\overline{A'B'}$.



The graph paper is a real-life example of a plane. When an object is moved, flipped, or rotated, it is the plane that the object is on that moved. This means that every point on the plane is moved, flipped, or rotated.

a. Draw $\overline{YZ}$ on the same plane as $\overline{AB}$. Then, draw $\overline{Y'Z'}$ on the second plane to show how $\overline{YZ}$ would move in the same way as $\overline{AB}$ when the plane is moved.
Answers will vary in the location of $\overline{YZ}$ which will also yield varying locations of $\overline{Y'Z'}$. Answers should represent a translation. Arrows (vectors/directed line segments) drawn between points Y and Y' should be the same length and go in the same direction as the arrows (vectors/directed line segments) drawn between Z and Z'.

Consider providing students with graph or patty paper in order to replicate what is presented in the narrative of question 1. This may provide a kinesthetic entry point to better see the manipulation of the image.

b. Draw the arrows from the endpoints of $\overline{YZ}$ to the endpoints of $\overline{Y'Z'}$. Compare the arrows from $\overline{YZ}$ to $\overline{Y'Z'}$ to the arrows from $\overline{AB}$ to $\overline{A'B'}$. What do you notice?
Answers will vary. I noticed that the arrows are parallel. Arrows drawn between points Y and Y' are the same length as the arrows drawn between Z and Z'. Arrows point in the same direction.

4.6.2 Exploration: Transforming Parts (continued)

Isometry is a transformation that preserves distance and angle measure.

c. Use the definition of isometry to justify that Traci's example is an isometry.
Answers will vary. I noticed the distance between points A and B is the same as the distance between points A' and B'. The distance between Y and Z is the same as the distance between Y' and Z'. $\overline{AB}$ is the same length as $\overline{A'B'}$. $\overline{YZ}$ is the same length as $\overline{Y'Z'}$.

"Isometry" will be a new term for students. Question 1c is included intentionally to give students opportunity to extend their working definitions of isometry by making sense of the concept through an example.

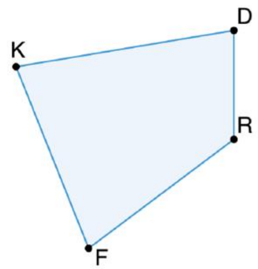
4.6.3 Exploration: Transformations and Isometry

Component Overview: *Students will be doing an exploration with transforming parts. For this activity, students should work with a partner or small group to go through each of the steps in the process of this activity. Students will use patty paper to go through each of the steps of this activity.*

4.6.3 Exploration: Transformations and Isometry

1. Quadrilateral $KDRF$ is shown.

   a. Use patty paper to copy the length of $\overline{RF}$. Label the endpoints on the patty paper. Flip the patty paper and place it in the space to the left of quadrilateral $KDRF$. Use the patty paper and any other tools to draw the copy of $\overline{RF}$. Name it $\overline{R'F'}$.



A ***transformation*** is a one-to-one correspondence between two sets of points.

   b. Work with your partner to complete the statement:

   Since there are a set of points on $\overline{RF}$ and a set of points on $\overline{R'F'}$, it is not enough to just state that the endpoints R and F correspond with endpoints R' and F' because a transformation is a one-to-one correspondence between two sets of points. Therefore, to show that $\overline{RF} \cong \overline{R'F'}$, all of the points on $\overline{RF}$ have to have a matching point on $\overline{R'F'}$.

Be mindful that this activity requires the use of patty paper to complete. Students may need additional support when it comes to using the patty paper to complete each step. Consider monitoring for opportunities or moments to briefly review student work so that observations or conclusions are more precise and accurate.

Students may struggle with what to use to complete the blanks. Guide students to refer back to the definition for a clue instead of giving them the answer. Often answers can be found within the content of the lesson. This strategy supports students in becoming more independent learners.

4.6.3 Exploration: Transformations and Isometry (continued)

c. Use the steps to apply the same transformations to each part of quadrilateral $KDRF$.

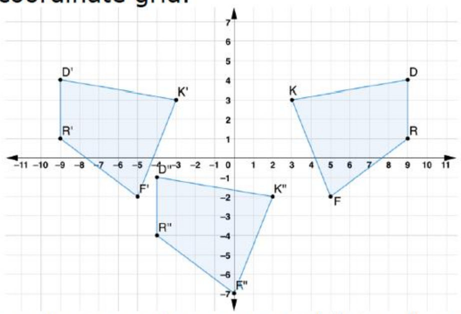
- Use patty paper to copy $\angle RFK$. Use the copy of $\angle RFK$ to create $\angle R'F'K'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KF}$. Name it $\overline{K'F'}$.
- Use patty paper to copy $\angle FRD$. Use the copy of $\angle FRD$ to create $\angle F'R'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{RD}$. Name it $\overline{R'D'}$.
- Use patty paper to copy $\angle FKD$. Use the copy of $\angle FKD$ to create $\angle F'K'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KD}$. Name it $\overline{K'D'}$.

d. Explain why when drawing all of the parts of the quadrilateral $KDRF$ to create quadrilateral $K'D'R'F'$, the movements of the patty paper were the same as when drawing $\overline{R'F'}$.
Answers will vary. The distance between all of the points on the quadrilateral remained the same as it did when making a copy of $\overline{RF}$ to draw $\overline{R'F'}$. To make a copy of quadrilateral $KDRF$, all parts of the new quadrilateral, $K'D'R'F'$, were placed in the same position with respect to each other as the parts of $KDRF$. All points in the new figure were the same distance from each other and placed in the same way, with respect to each other, when making the copy. All the angle measures are the same. Note that quadrilateral $KDRF$ is the preimage, and quadrilateral $K'D'R'F'$is the image.

Some students will learn more effectively during question 1c by individually engaging with each step, while others will work better articulating and verifying their steps with a partner.

If students are unsure how to answer question 1d, consider using the following questions as prompts:
- *Why do you think the distances between all of the points on the quadrilateral remained the same as they did during making a copy of RF to draw R'F'?*
- *Why do all angles stay the same in this case?*
- *Which of the quadrilaterals is the preimage and which is the actual image? Does it matter?*

2. Quadrilaterals $KDRF$, $K'D'R'F'$, and $K''D''R''F''$ are shown on the coordinate grid.



   a. Use patty paper to copy quadrilateral $KDRF$.
   b. Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $KDRF$.

   c. Describe what could be done so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K'D'R'F'$.
   Answers will vary. Reflect quadrilateral $KDRF$ across the y-axis.
   Reflect quadrilateral $KDRF$ using the line of reflection of $x = 0$.
   A horizontal translation of quadrilateral $KDRF$ left 6 units and a reflection with the line of reflection $x = -3$.

   d. Was the movement of the patty paper an example of isometry?
   Yes. When the copy of quadrilateral $KDRF$ was placed on top of quadrilateral $K'D'R'F'$, so that the corresponding points were on top of each other, the distance between the points on quadrilateral $K'D'R'F'$ was the same as the distance between the points on quadrilateral $KDRF$. The angle measures were the same.

   e. Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $K'D'R'F'$.

When reviewing question 2, consider allowing a student (or student pair) to come to the board or document camera to show the steps from question 2a and 2b.

During the review of question 2c, engage students in a discussion where they are using accurate translation language when completing this process.

Monitor opportunities for more precise academic language in moments when students are describing things like “reflect across the y-axis” as “flip up” or similar informal vocabulary.

Question 2c allows for multiple entry points, as students may see transformed images differently. Allow students to describe the transformations based on what they see and not a predetermined set of moves.

4.6.3 Exploration: Transformations and Isometry (continued)

f. Describe what should be done next so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$.
A vertical translation down 5 units and a horizontal translation right 5 units.

g. Was the movement of the patty paper an example of isometry?
Yes. When the copy of quadrilateral $KDRF$ was placed on top of quadrilateral $K''D''R''F''$, the distance between the points on quadrilateral $K''D''R''F''$ was the same as the distance between the points on quadrilateral $K'D'R'F'$. The angle measures were the same.

h. Determine if the copy of quadrilateral $KDRF$ could be moved with just one transformation so that it lays on top of quadrilateral $K''D''R''F''$.
No. You would need to perform more than one transformation in order to lay a copy of quadrilateral $KDRF$ on top of quadrilateral $K''D''R''F''$.

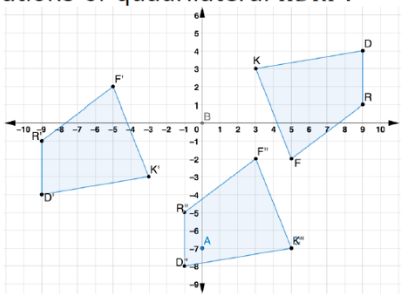
3. There is at least one other sequence of transformations that can be used to move the copy of quadrilateral $KDRF$ so that it lays on top of quadrilateral $K''D''R''F''$. Work with your partner to describe one of these sequences of transformations.
Answers will vary. However, the answer to this question depends on the initial placement of the copy of quadrilateral $KDRF$ when performing the first transformation. For instance, to move a copy of quadrilateral $KDRF$ from quadrilateral $KDRF$ to quadrilateral $K''D''R''F''$, performing a transformation of a reflection across $x = 3$ would then require a vertical translation of 5 units down, and horizontal translation of 1 unit left. A different initial placement could be a reflection across $x = 2$, which would then require a vertical translation of 5 units down, and horizontal translation of 1 unit right.

For students working in pairs or groups and stuck on question 2g or 2h, consider the following questions:
- *What conditions did this transformation meet in order to be an example of isometry?*
- *Why can't a copy of this be moved with just one transformation?*

Student responses to question 3 will depend on the initial placement of the copy of the quadrilateral. Monitor for varying responses or approaches to later highlight when reviewing in a whole-group setting.

4.6.3 Exploration: Transformations and Isometry (continued)

4. The coordinate grid shows a different sequence of transformations of quadrilateral $KDRF$.



Describe what movements should be done with your patty paper copy of quadrilateral $KDRF$ so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$, by first laying on top of quadrilateral $K'D'R'F'$.

Answers will vary. A rotation counterclockwise $180°$ with the center of rotation at the point $(0,0)$ then a vertical translation of 4 units down and a horizontal translation of 8 units right.

A reflection across the y-axis, a reflection across the x-axis, then a vertical translation 4 units down and a horizontal translation 8 units right.

Monitor students as they close their exploration, using questions like the following to gauge their conceptual understanding.

- *How is this sequence of transformations different than the prior?*
- *Does this sequence of transformations represent isometry? Why or why not?*

Consider having students work on question 4 in pairs and then find another pair of students to compare answers with and discuss.

4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

Component Overview: *Students work to formalize the definition of congruence in terms of rigid motions based on the Exploration in the prior activity (4.6.3).*

4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

1. For both sequences of transformations of quadrilateral $KDRF$, the paper copy of quadrilateral $KDRF$ was

   - ○ congruent
   - ○ not congruent

   to

   - ○ $K'D'R'F'$ only.
   - ○ $K''D''R''F''$ only.
   - ○ both $K'D'R'F'$ and $K''D''R''F''$.

   The transformations used in both sequences were

   - ○ dilations.
   - ○ rigid motions.
   - ○ dilations and rigid motions.

   All of the points of

   - ○ $K'D'R'F'$
   - ○ $K''D''R''F''$
   - ○ $K'D'R'F'$ and $K''D''R''F''$

   can be mapped to all of the points of $KDRF$.

Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

2. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $KDRF$ resulted in quadrilateral $KDRF \cong$ quadrilateral $K''D''R''F''$.
   In both sequences of transformations only rigid motions (translations, reflections, or rotations) were used to map KDRF onto K″D″R″F″ so the figures are congruent by the definition of congruence in terms of rigid motions

Synthesis is a key component for strategies of Visible Learning (see Unit Guide). Students approach synthesis differently. Consider routines like Three Reads, Think-Pair-Share, or Stronger and Clearer to give students multiple entry points for synthesis in this activity.

Multiselect questions are often seen on high-stakes assessments. Consider a Think Aloud to model strategies or approaches for answering these questions.

For example, begin with the word "congruent" and move to the next word bank to determine a choice that makes sense with congruent, then move forward to each word bank to see if a chain of correct responses can be made to make one correct answer.

4.6.5 Wrap-Up: Growth Mindset

Component Overview: *Students will complete a growth mindset survey related to the concepts covered in this lesson.*

4.6.5 Wrap-Up: Growth Mindset

The power of "YET."

"The power of believing that you can improve." - Carol Dweck

Select all of the concepts that apply to your understanding right now.

Answers will vary.

1. I don't understand
   - isometry
   - transformations
   - the definition of congruence in terms of rigid motions

   YET.

2.
   - Isometry
   - Mapping
   - One-to-one correspondence
   - Definition of congruence in terms of rigid motions

   doesn't make sense to me YET.

3. I can explain
   - isometry.
   - mapping.
   - one-to-one correspondence.
   - transformations.
   - the definition of congruence in terms of rigid motions.

Lesson 7 incorporates a video on growth mindset during the Warm-Up. Consider prompting students to connect their responses here to their working definition of growth mindset and what that can look and sound like for them in this unit.